Load Frequency Control

Load frequency control :

f_{rated} = 50 Hz (Ideal) ; f_{range} = 49.9 to 50.05 Hz (practical)

Disadvantages of frequency deviation :

(1) X'mer / CT / PT core saturated ($\phi_m \propto \frac{1}{f}$)

(2) Speeds of motors are affected.

(3) Hysteresis losses are reduced

(4) Turbine blades are damaged.

(5) Performance of thermal plants are affected.

Methods of frequency control :

(1) Primary control - speed governing valve control.

(2) Secondary control

(a) Running of peak load plants like pumped storage, hydro.

(b) Load regulation using AUFR [Automatic under frequency relays]

(1) Speed Governing valve control :

Fly ball Governor is used to control the frequency within the safe limits.

Speed changer will bring the frequency from the deviated value to the rated value of 50 Hz.

Frequency Deviation (Δf) occurs due to $\Delta (P_g - P_d)$

P_g = Active power Generation ; P_d = Active power demand

$\Delta (P_g - P_d)$ causes change in :

(1) Change in K.E of Gen

(2) $B = \frac{\partial P_D}{\partial f} = \frac{1}{K_p}$ = Damping Torque coefficient

$$B = \frac{\%\ \text{of power change} \times \text{Total power}}{\%\ \text{of frequency change} \times \text{Rated freq.}}$$

(3) Tie line power flow

Steady state frequency Deviation (Δf)

* $\Delta f = (f_n - f_i)$ Hz ; f_n = New freq. Hz ; f_i = Initial / Rated frequency Hz = 50 Hz

H = Inertia constant of Generator MW sec / MVA ; S = Rating of Generator (MVA)

W = Kinetic Energy of Generator

$$W = H \cdot S = \frac{1}{2} I \omega^2$$ MW sec ; \omega=2\pi f

T_d = Time delay in Governing system in sec. ; ΔP_d = load demand (or) load loss in MW.

$f_i \propto (H \cdot S)^{1/2}$

For load loss : $f_n \propto (H \cdot S + \Delta P_D \cdot T_d)^{1/2}$

For load demand : $f_n \propto (H \cdot S - \Delta P_D \cdot T_d)^{1/2}$

Additional load demand :

$$f_n = f_i \left(\frac{HS - \Delta P_D \cdot T_d}{HS} \right)^{1/2}$$

Load loss / Trip of T/L :

$$f_n = f_i \left(\frac{HS + \Delta P_D \cdot T_d}{HS} \right)^{1/2}$$

Speed Regulation parameter

$$R = -\frac{\Delta f}{\Delta P}$$

$p.u = \text{per unit} = \frac{\text{Actual quantity}}{\text{Base quantity}}$; f_{rated} -> Base frequency ; S -> Base MVA

$$R = \frac{-\Delta f / f_{rated}}{\Delta P / S} = -\frac{\Delta f}{\Delta P} \times \frac{S}{f_{rated}}$$

Assume $\Delta P = S$:

$$R_{pu} = -\frac{\Delta f}{f_{rated}} \quad p.u$$

Eg: $\Delta f = f_{NL} - f_{FL} = 2$ Hz ; $\Delta P = P_{FL} - P_{NL} = 100$ MW ; $S = 100$ MVA ; $f_{rated} = 50$ Hz

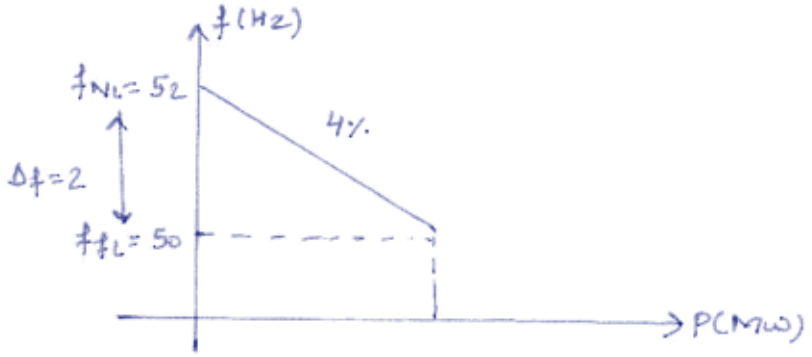
$$R = -\frac{\Delta f}{\Delta P} = -\frac{2}{100} = -0.02 \text{ Hz/MW}$$

$$R_{pu} = -\frac{\Delta f}{f_{rated}} = -\frac{2}{50} = -0.04 \text{ p.u} \; ; \; |R_{pu}| = 0.04 \text{ p.u}$$

$$\%R = R_{pu} \times 100 = 0.04 \times 100 = 4\ \%$$

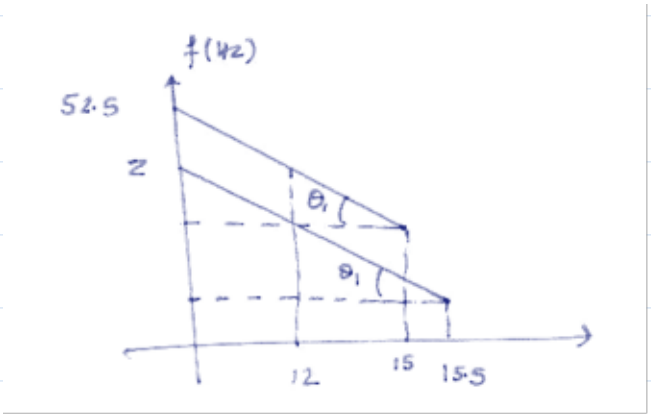
$$\Delta f = (\%R) \times f_{rated}$$

$$= \frac{4}{100} \times 50 = 2 \text{ Hz}$$



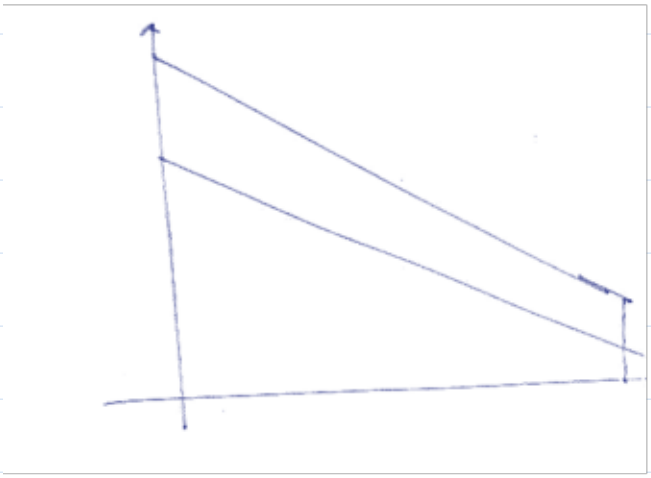
Droop characteristic: f (Hz) against P (MW), no-load frequency 52 Hz falling 4% to full-load 50 Hz, delta f = 2 Hz

DAY # 9 [17/9/25] :

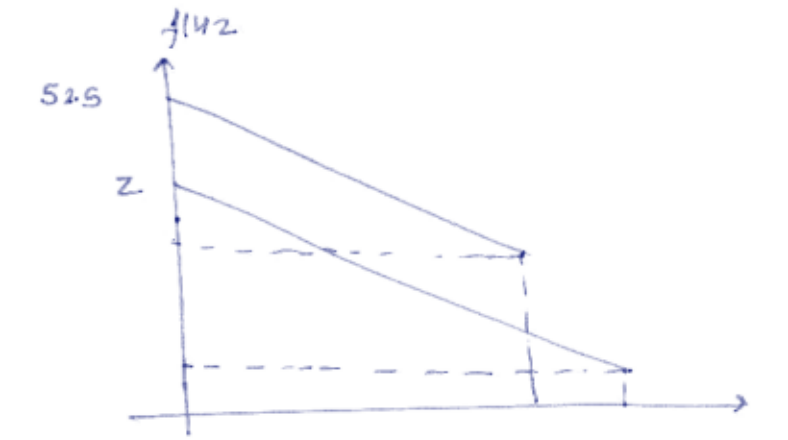


Two parallel droop lines: f (Hz) from 52.5 and 52 against P, with 12, 15 and 15.5 marked, angles theta1

$$K = \frac{2.5}{15} = 0.166$$

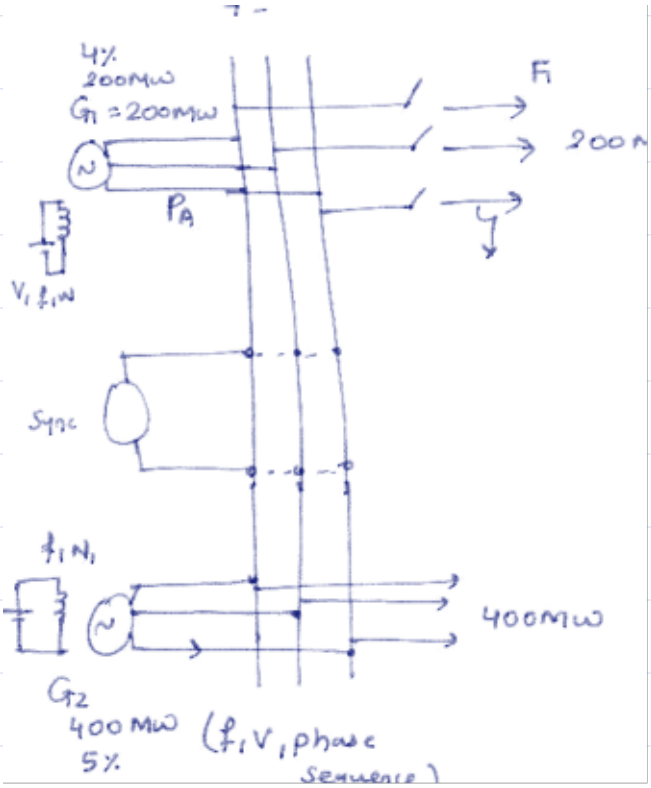


Two droop lines of the same slope shifted by speed changer setting

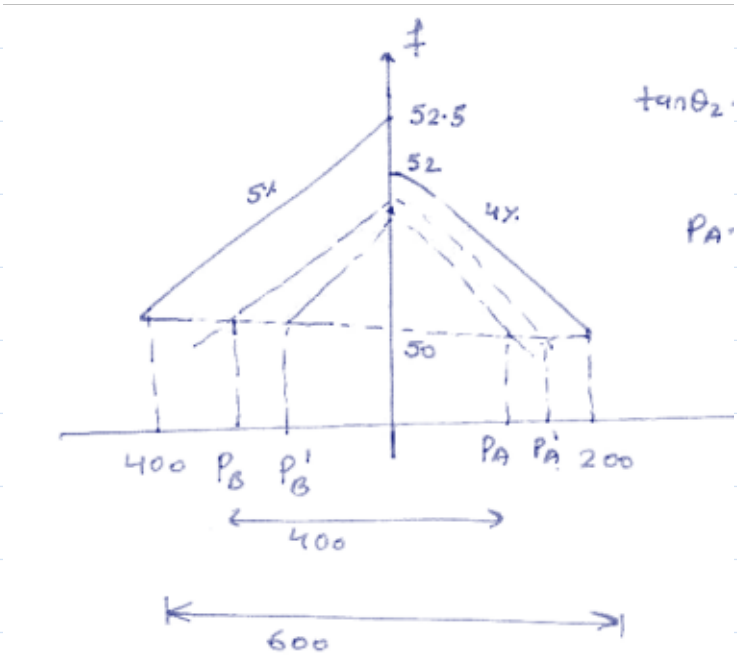


Two droop lines (52.5 and 52 Hz intercepts) with operating points marked by dashed lines

Load sharing of parallel Generators :



Generator G1 = 200 MW, 4 % droop, feeding feeder F1 (200 MW) through the bus; synchronising machine; Generator G2 = 400 MW, 5 % droop (f, V, phase sequence matched) feeding 400 MW



f against P: G2 line 5 % from 52.5 Hz, G1 line 4 % from 52 Hz, common operating frequency 50 Hz, loads PB, PB' and PA, PA', capacity 400 and 200, total 600

$$\tan\theta_1 = \frac{52 - f_n}{P_A} = \frac{52 - 50}{200} = \frac{2}{200} \quad -(1)$$

$$\tan\theta_2 = \frac{52.5 - f_n}{P_B} = \frac{52.5 - 50}{400}$$

$P_A + P_B = 400 \Rightarrow P_B = 400 - P_A$

$$\frac{52.5 - f_n}{400 - P_A} = \frac{2.5}{400}$$

$f_n = 50.77\ Hz\ ;\ P_A = 123\ MW\ ;\ P_B = 277\ MW$

Before tripping of feeder-1 :

$$\frac{P_1}{P_2} = \frac{200}{400} = \frac{1}{2} = 0.5$$

After tripping of F1 :

$$\frac{P_A}{P_B} = \frac{123}{277} = 0.44$$

(2) For economic load Dispatch i.e. to minimize overall cost of Gen, load should be shared in proportion to the Gen capacity.

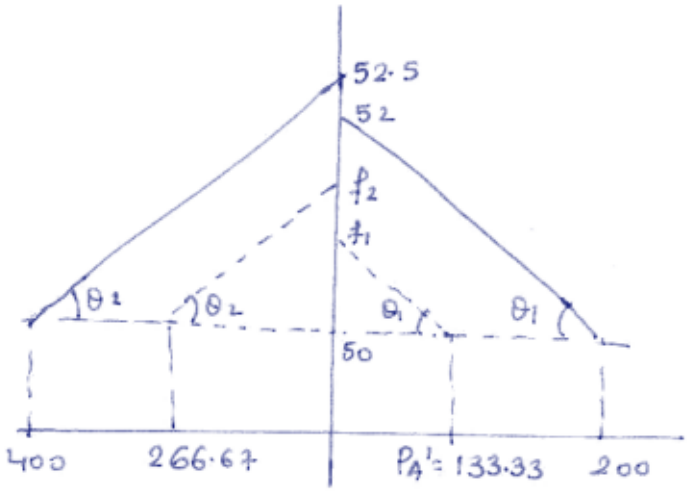
$$\frac{P'_A}{P'_B} = \frac{P_1}{P_2} = 0.5$$

$P'_A + P'_B = 400\ ;\ P'_B + 0.5P'_B = 400 \rightarrow P'_B = 266.67\ MW\ ;\ P'_A = 133.33\ MW$

No load frequencies :

$$\tan\theta_1 = \frac{f_1 - 50}{133.33} = \frac{52 - 50}{200} \rightarrow f_1 = 51.33\ Hz$$

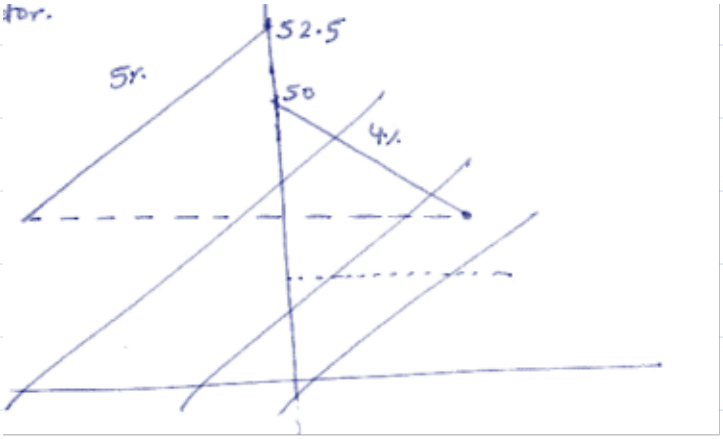
$$\tan\theta_2 = \frac{f_2 - 50}{266.67} = \frac{52.5 - 50}{400} \rightarrow f_2 = 51.66\ Hz$$



Droop lines with no-load frequencies f1, f2 shifted so loads share 266.67 and 133.33 MW at 50 Hz

Eg: Two Generators G_1 and G_2 each having a capacity of 200 MVA operating at no load freq 50 Hz; G_1 has 4 % drooping, G_2 has 5 % drooping ch. For a load of 300 MW find load sharing of G_1 and G_2 and the new operating freq.

(ii) What is the maximum load that can be shared by both the Generators without overloading any of the Generators.



Droop lines for G1 (4 %) and G2 (5 %) from 50 Hz; shifted lines illustrating load sharing

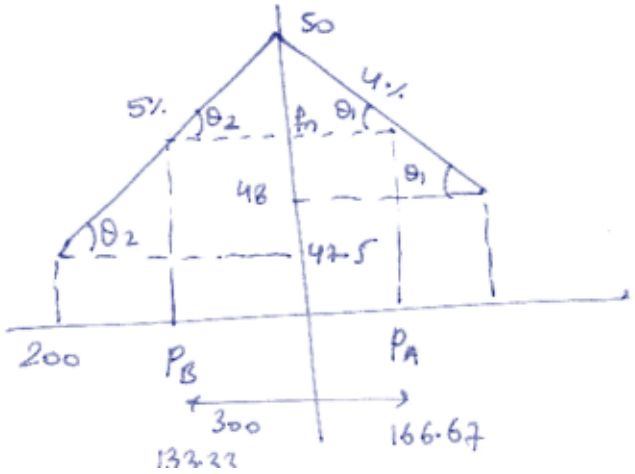
$$\tan\theta = \frac{52 - f_n}{P_A}$$

$$\tan\theta_1 = \frac{50 - f_n}{P_A} = \frac{50 - 48}{200} \quad -(1)$$

$$\tan\theta_2 = \frac{50 - 47.5}{200} = \frac{50 - f_n}{P_B} = \frac{50 - f_n}{300 - P_A}$$

$P_A = 166.67$ MW

$f_n = 48.34$ Hz



f against P with 50 Hz no-load for both machines: 5 % line and 4 % line meet at common frequency 48 Hz; G2 at 42.5 Hz at 200 MW; PB and PA with total 300

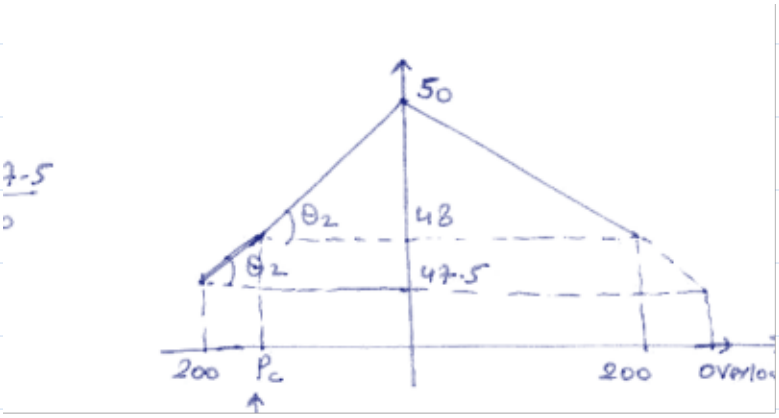
$$\tan\theta_1 = \frac{50 - f_n}{P_A} = \frac{50 - 48}{200} \quad -(1)$$

$$\tan\theta_2 = \frac{50 - 42.5}{200} = \frac{50 - f_n}{P_B} = \frac{50 - f_n}{300 - P_A}$$

$P_A = 166.67\ MW\ ;\ f_n = 48.34\ Hz\ ;\ P_B = 133.33\ MW$

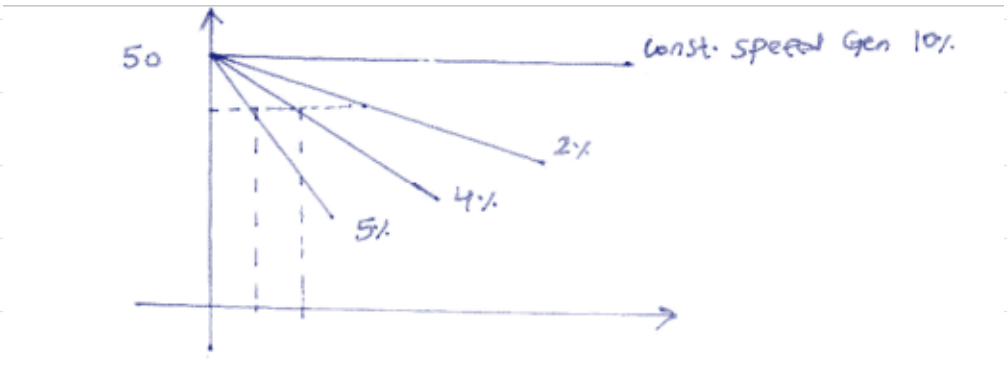
$$\tan\theta_2 = \frac{50-48}{P_C} = \frac{50-47.5}{200} \Rightarrow P_C = 160\ MW$$

$P_{max} = 200 + 160 = 360\ MW$ @ 48 Hz (G1 at full 200 MW, G2 at 160 MW)



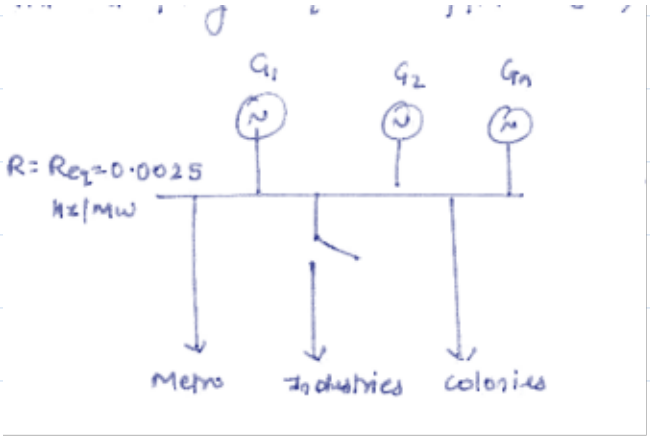
Droop lines: G2 at 48 Hz carries PC = 160 MW (under load), G1 at 200 MW; overload region beyond; Pmax

* In case of parallel operation of Generators, the Gen which has lowest Drooping ch. will share maximum load.
* If const. speed or 0 % Drooping ch. Generator is connected with the other Generators having 2, 4, 6 % etc. drooping ch., then 0 % drooping ch. generator will share entire load change without loading other parallel Generators, assuming it has infinite capacity.



f against P from 50 Hz: const. speed generator (0 %) is a flat line; drooping lines 2 %, 4 %, 5 %

Damping Torque coefficient (B)



Generators G1, G2, Gn on a common bus feeding loads: Metro, Industries, Colonies; R_eq = 0.0025 Hz/MW

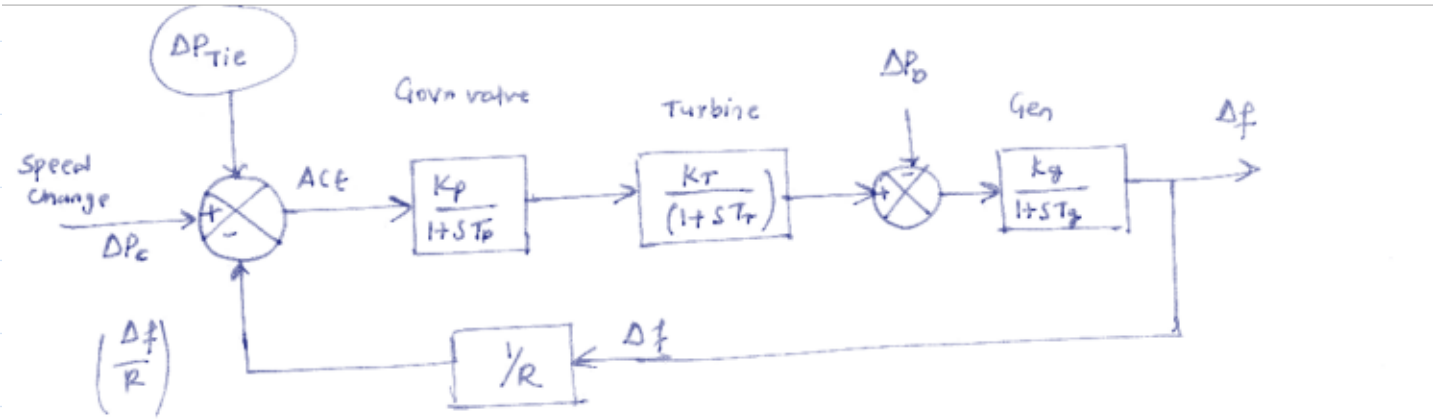
$$B = \frac{\%\ \text{power change} \times \text{Total power}}{\%\ \text{frequency change} \times f_{rated}}$$

$$(1)\ R = -\frac{\Delta f}{\Delta P_D}\ ;\ \Delta f = -\Delta P_D \cdot R = -\frac{\Delta P_D}{(1/R)}$$

ΔP_D = -ve for load loss or trip ; = +ve for load demand.

$$\Delta f = -\frac{\Delta P_D}{\left(B + \frac{1}{R}\right)}, \qquad \beta = B + \frac{1}{R} = AFRC$$

Block Diagram Representation of single Area :



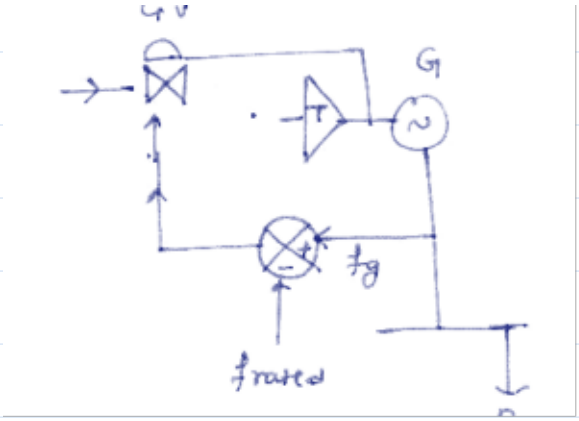
Block diagram: speed change delta Pc and tie-line power delta PTie summed to give ACE; governor Kp/(1+sTp); turbine Kt/(1+sTt); load change delta PD subtracted; generator Kg/(1+sTg) gives delta f; feedback 1/R of delta f

ACE = Area control error

$$ACE = \Delta P_c + \Delta P_{Tie} - \left(\frac{\Delta f}{R}\right)$$

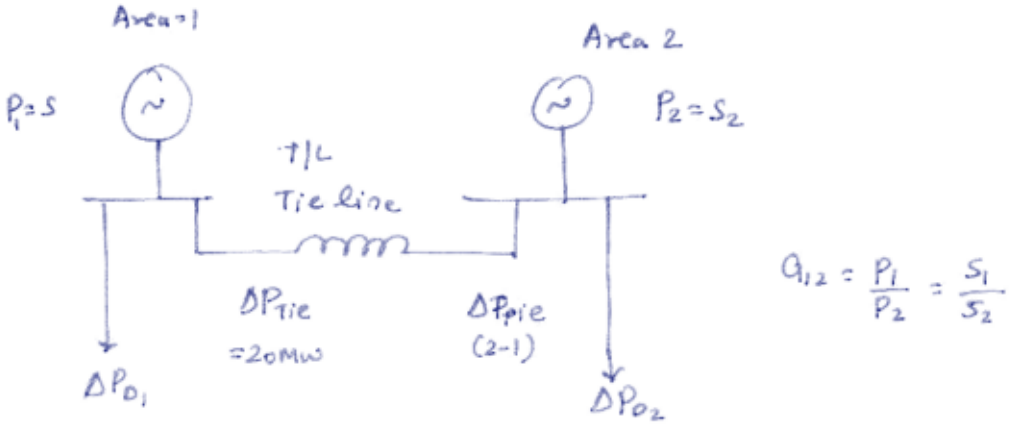
For ΔP_c = const. , b = -1/R : $ACE = \Delta P_{Tie} + b(\Delta f)$

K_p, K_g, K_T = Gain const. ; T_p, T_g, T_T = Time const. ; R = speed Reg. parameter ; ΔP_{Tie} = Tie line power loss.



Governor valve GV, turbine T, generator G with frequency feedback fg compared against f rated; bus with load PD

Two Area frequency control



Area 1 (P1 = S1) and Area 2 (P2 = S2) joined by T/L tie line, delta PTie = 20 MW, loads delta PD1 and delta PD2; a12 = P1/P2 = S1/S2

$$\Delta f = -\frac{(\Delta P_{D2} + a_{12}\Delta P_{D1})}{(\beta_2 + a_{12}\beta_1)}$$

$$\Delta P_{Tie(1-2)} = \frac{\Delta P_{D1}\beta_2 - \Delta P_{D2}\beta_1}{(\beta_2 + a_{12}\beta_1)} = -a_{12}\Delta P_{Tie(1-2)}$$

where Area Frequency Response ch. = AFRC : $\beta_1 = B_1 + \frac{1}{R_1}$; $\beta_2 = B_2 + \frac{1}{R_2}$

Eg: You are given 2 system areas connected by a tie line with the following data :

Area 1 : $R_1 = 0.01$ p.u ; $B_1 = 0.8$ p.u ; Base MVA = 500

Area 2 : $R_2 = 0.02$ p.u ; $B_2 = 1.0$ p.u ; Base MVA = 500

A load demand change of 100 MW occurs in area 1. What is the steady state frequency and what is the change in tie line flow? Assume both areas were working at a nominal freq of 50 Hz to begin.

$\Delta P_{D1} = \frac{100}{500} = 0.2$ p.u ; $\beta_2 = 1 + \frac{1}{0.02} = 51$; $\beta_1 = 0.8 + \frac{1}{0.01} = 100.8$

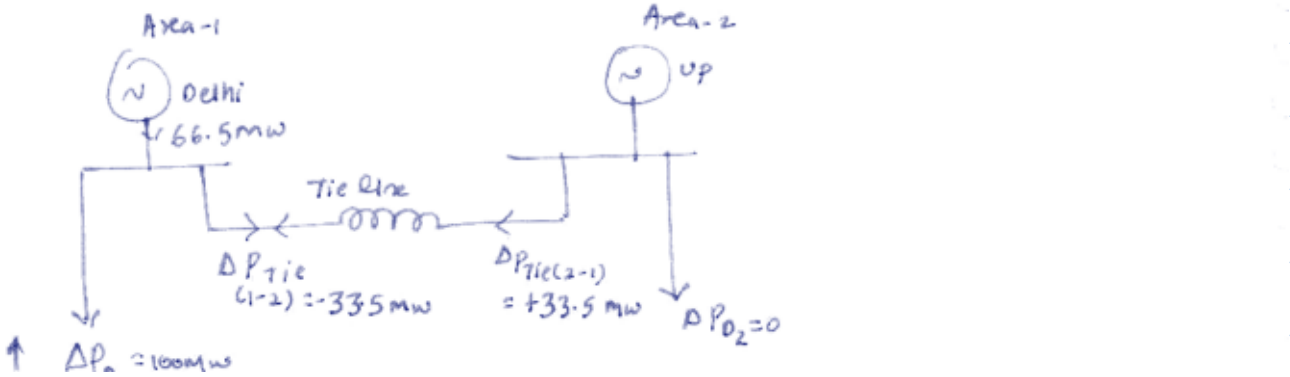
$$\Delta f = -\frac{(\Delta P_{D2} + a_{12}\Delta P_{D1})}{(\beta_2 + a_{12}\beta_1)} = -\frac{(0 + 1 \times 0.2)}{(51 + 100.8)} = -1.3 \times 10^{-3} \text{ p.u}$$

$\Delta f = -1.3 \times 10^{-3} \times 50 = -0.065$ Hz ; $f_n = 50 - 0.065 = 49.935$ Hz ; $a_{12} = \frac{S_1}{S_2} = \frac{500}{500} = 1$

Tie line power flow :

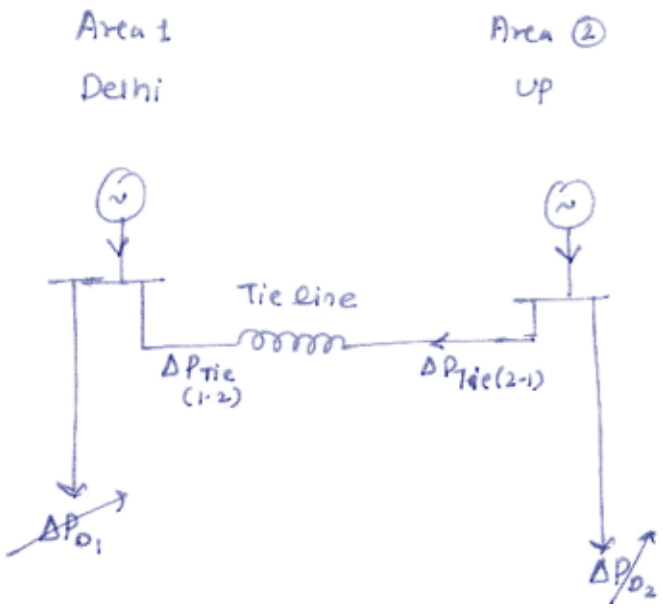
$$\Delta P_{Tie(1-2)} = \frac{(\beta_1 \Delta P_{D2} - \beta_2 \Delta P_{D1})}{(\beta_2 + a_{12}\beta_1)} = \frac{-51 \times 0.2}{(51 + 1 \times 100.8)} = -0.067 \text{ p.u}$$

$\Delta P_{Tie(1-2)} = -0.067 \times 500 = -33.5$ MW ; $\Delta P_{Tie(2-1)} = -a_{12}\Delta P_{Tie(1-2)} = 33.5$ MW.



Area 1 (Delhi) with load increase of 100 MW, supplying 66.5 MW itself; tie line carrying 33.5 MW from Area 2 (UP); delta PD2 = 0

Methods of 2 Area frequency control :



Area 1 (Delhi) and Area 2 (UP) joined by a tie line; load changes delta PD1 and delta PD2; tie power delta PTie(1-2), delta PTie(2-1)

(1) Flat frequency control : Load changes of Area 2 is supplied by the Area 1 generator for maintaining constant frequency in both the areas, but in this method tie lines may be overloaded causing interruption to the 2 areas due to isolation.

(2) Flat Tie Bias / Tie line control : In this method control signal is proportional to the changes in the tie line power and frequency variation, maintaining the constant power flow without overloading the tie line.

(3) Parallel frequency control : Load changes of either busbar A or B (Area 1 or Area 2) are simultaneously supplied by both generators for maintaining frequency within the range without overloading of tie line.

Load Regulation :

At various stages for maintaining the grid frequency.

AUFR stages : Stage 1 : 49.4 Hz ; Stage 2 : 49.2 Hz ; Stage 3 : 49 Hz ; Stage 4 : 48.8 Hz

$f + \frac{df}{dt}$ stages : Stage 1 : 49.5 - 0.3 Hz/sec ; Stage 2 : 49.3 - 0.2 Hz/sec

AUFR - Automatic under frequency Relay.

Travelling waves and Transients in the T/L :

* Power system N/W consisting of Generator, X'mer, T/L, busbars, electrical loads like motors etc. All these equipments withstand certain vtg / power / current withstanding capability.

* If any sudden changes in these parameters occur due to lightning surges, outages of equipment will induce large vtg and current in the system which will damage the electrical equipment. To protect these equipments, protecting equipment like lightning arrester, arcing horn, earth wire, counter poise etc. are installed in the power system N/W. For designing and identifying optimum locations, transient study analysis is conducted.